

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	264852	49.244832
1	[Git, Change, Create]	99583	18.515805
2	[Git, Change, Delete]	61316	11.400692
3	[Git, Change, Rename]	24820	4.614867
4	[File, Git]	16609	3.088168
5	[Git, Commit]	16390	3.047448
6	[Java, ByteCode, Parameter]	9457	1.758372
7	[Java, ByteCode, Member, Method]	9076	1.687531
8	[Java, ByteCode, Bound]	6545	1.216934
9	[Java, Value, ByteCode, Annotation]	5562	1.034162
10	[Java, ByteCode, Bound, ParameterizedType]	5029	0.935059
11	[Xml, Element]	2512	0.467065
12	[Java, ByteCode, Member, Field]	2238	0.416119
13	[Xml, Text]	1692	0.314599
14	[Java, ByteCode, Member, Constructor, Method]	1592	0.296006
15	[Java, ByteCode, Bound, WildcardType]	1079	0.200622
16	[Java, ByteCode, Bound, TypeVariable]	868	0.161390
17	[Java, ByteCode, Member, Method, Lambda]	839	0.155998
18	[Type, File, Java, ByteCode, JavaType]	698	0.129782
19	[Type, File, Java, ByteCode, Class]	675	0.125505
20	[Git, Change, Copy]	546	0.101520
21	[Json, Key]	523	0.097243
22	[Value, Json, Scalar]	513	0.095384
23	[Java, ByteCode, GenericDeclaration, Member, M...]	481	0.089434
24	[Type, File, Java, ByteCode, ResolvedDuplicate...]	477	0.088690
25	[Java, Value, ByteCode, Primitive]	364	0.067680
26	[Type, File, Java, ByteCode, ExternalType]	330	0.061358
27	[Author, Git, Person]	307	0.057082
28	[Committer, Git, Person]	248	0.046111
29	[Maven, Dependency]	205	0.038116

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

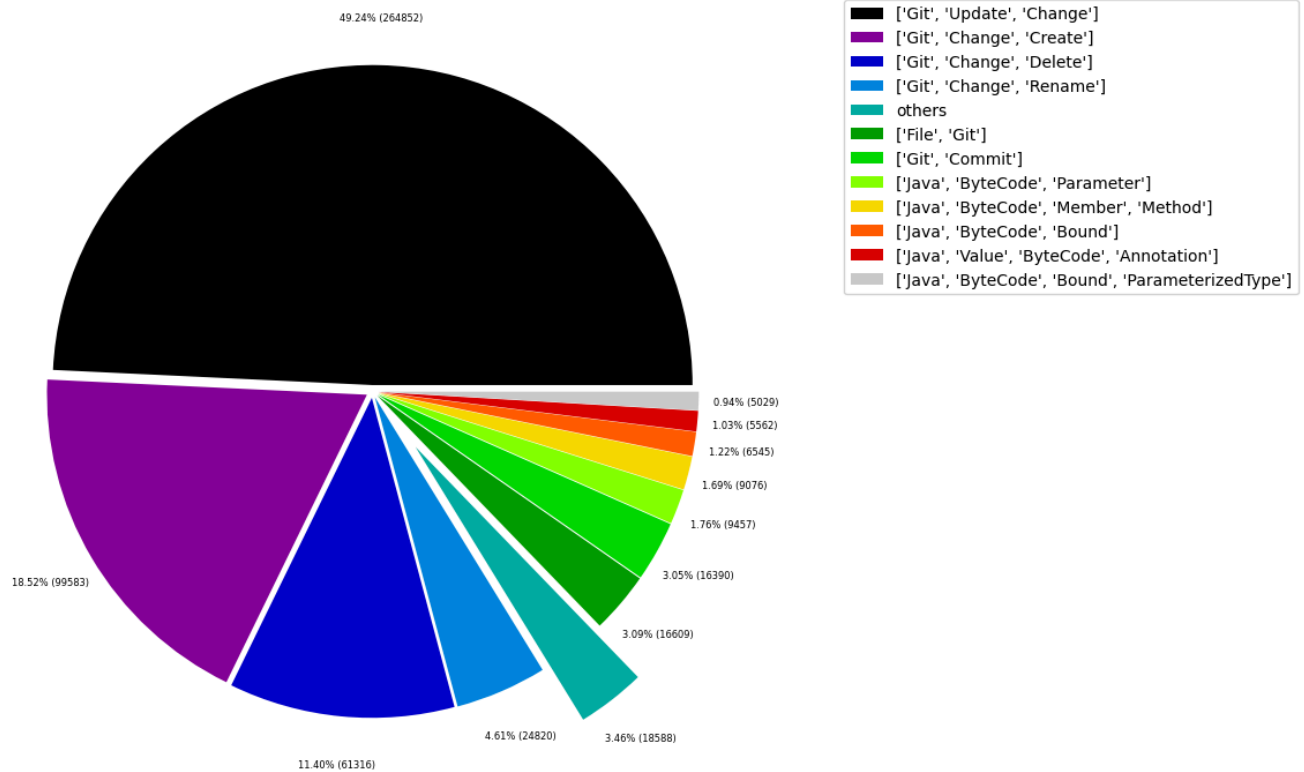


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000186
1	[Git, Branch]	1	0.000186
2	[Repository, File, Git]	1	0.000186
3	[File, Json]	2	0.000372
4	[File]	4	0.000744
5	[Type, File, Java, ByteCode, GenericDeclaratio...	4	0.000744
6	[Maven, Exclusion]	6	0.001116
7	[Value, Array, Json]	8	0.001487
8	[Java, ByteCode, GenericDeclaration, Member, C...	9	0.001673
9	[Type, File, Java, ByteCode, Throwable, Extern...	10	0.001859
10	[Type, File, Java, ByteCode, Throwable, Resolv...	11	0.002045
11	[Maven, Scm]	11	0.002045
12	[File, Maven, Xml, Pom, Document]	11	0.002045
13	[Type, File, Java, ByteCode, Void]	11	0.002045
14	[Java, ManifestSection]	11	0.002045
15	[File, Java, Manifest]	11	0.002045
16	[File, Artifact, Jar, Archive, Zip, Java]	11	0.002045
17	[File, Java, ServiceLoader]	13	0.002417
18	[Maven, PluginExecution]	13	0.002417
19	[File, Java, Properties]	13	0.002417
20	[Maven, ExecutionGoal]	13	0.002417
21	[Type, File, Java, ByteCode, Enum]	17	0.003161
22	[jQAssistant, Rule, Concept]	19	0.003533
23	[Maven, Configuration]	20	0.003719
24	[Maven, Plugin]	20	0.003719
25	[Xml, Attribute]	22	0.004091
26	[Type, File, Java, ByteCode, PrimitiveType]	39	0.007251
27	[Type, File, Java, ByteCode, Class, Throwable]	42	0.007809
28	[Type, File, Java, ByteCode, Annotation]	42	0.007809
29	[Xml, Namespace]	44	0.008181

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

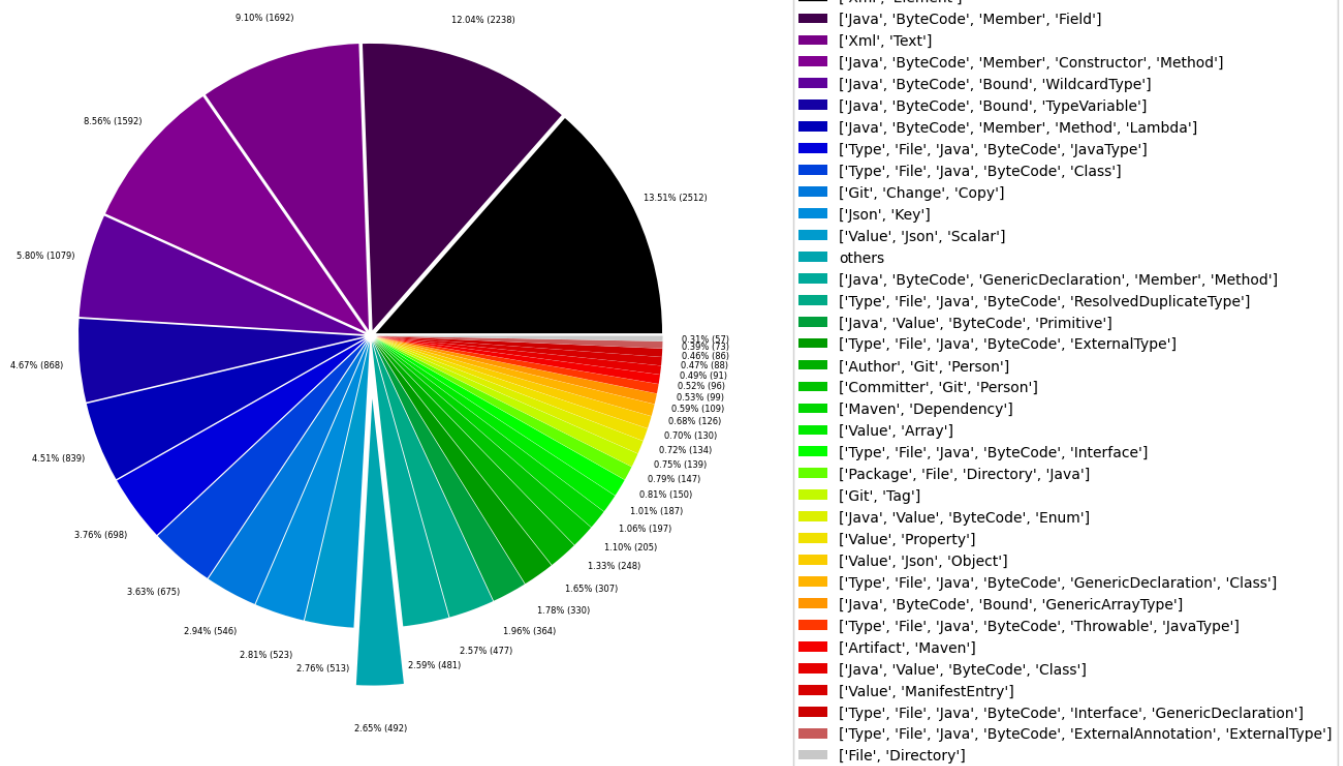


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

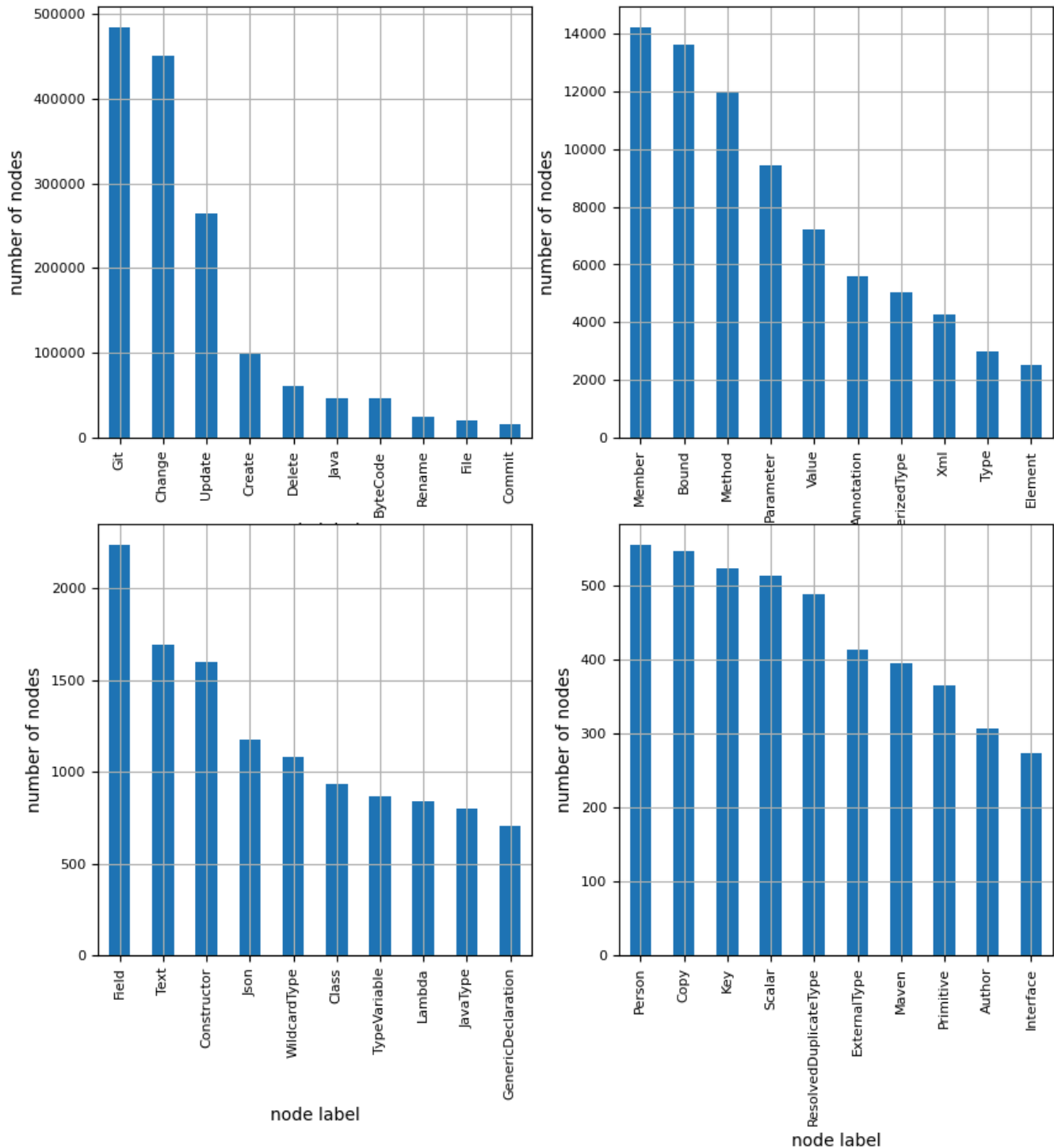
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	484820	90.144229
1	Change	451117	83.877715
2	Update	264852	49.244832
3	Create	99583	18.515805
4	Delete	61316	11.400692
5	Java	46666	8.676768
6	ByteCode	46457	8.637908
7	Rename	24820	4.614867
8	File	19861	3.692823
9	Commit	16390	3.047448
10	Member	14235	2.646762
11	Bound	13630	2.534272
12	Method	11997	2.230643
13	Parameter	9457	1.758372
14	Value	7226	1.343555
15	Annotation	5604	1.041971
16	ParameterizedType	5029	0.935059
17	Xml	4281	0.795981
18	Type	2979	0.553896
19	Element	2512	0.467065
20	Field	2238	0.416119
21	Text	1692	0.314599
22	Constructor	1601	0.297679
23	Json	1176	0.218658
24	WildcardType	1079	0.200622
25	Class	934	0.173662
26	TypeVariable	868	0.161390
27	Lambda	839	0.155998
28	JavaType	797	0.148189
29	GenericDeclaration	706	0.131269
30	Person	555	0.103193
31	Copy	546	0.101520
32	Key	523	0.097243
33	Scalar	513	0.095384
34	ResolvedDuplicateType	488	0.090735
35	ExternalType	413	0.076790
36	Maven	395	0.073444
37	Primitive	364	0.067680
38	Author	307	0.057082
39	Interface	273	0.050760

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1668574

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	451117	27.036080
1	MODIFIES	451117	27.036080
2	UPDATES	264852	15.872955
3	CREATES	124949	7.488370
4	DELETES	86136	5.162252
5	COMMITTED	32780	1.964552
6	RENAMES	24820	1.487498
7	INVOKES	22756	1.363799
8	HAS_PARENT	19906	1.192995
9	OF_TYPE	19011	1.139356
10	DEPENDS_ON	18178	1.089433
11	HAS_FILE	16609	0.995401
12	HAS_COMMIT	16390	0.982276
13	DECLARES	14589	0.874339
14	OF_RAW_TYPE	13531	0.810932
15	HAS	10124	0.606746
16	HAS_NEW_NAME	9599	0.575282
17	RETURNS	8301	0.497491
18	READS	6113	0.366361
19	HAS_ACTUAL_TYPE_ARGUMENT	5936	0.355753
20	ANNOTATED_BY	5561	0.333279
21	CHANGED_TOGETHER_WITH	4683	0.280659
22	OF_GENERIC_TYPE	4539	0.272029
23	SIMILAR	3496	0.209520
24	CONTAINS	3309	0.198313
25	RESOLVES_TO	2698	0.161695
26	RETURNS_GENERIC	2679	0.160556
27	HAS_FIRST_CHILD	2512	0.150548
28	HAS_LAST_CHILD	2512	0.150548
29	HAS_ELEMENT	2490	0.149229

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

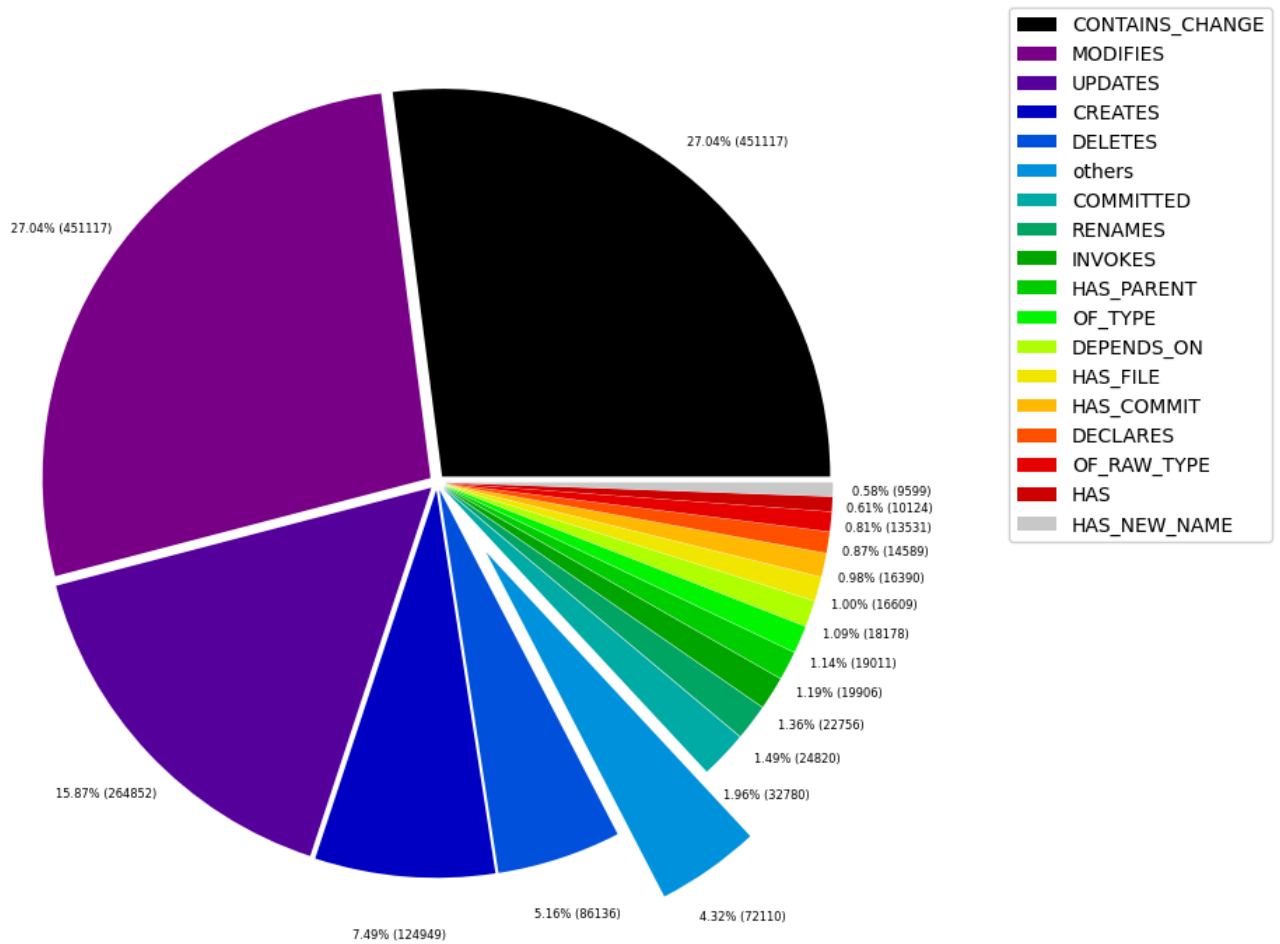


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

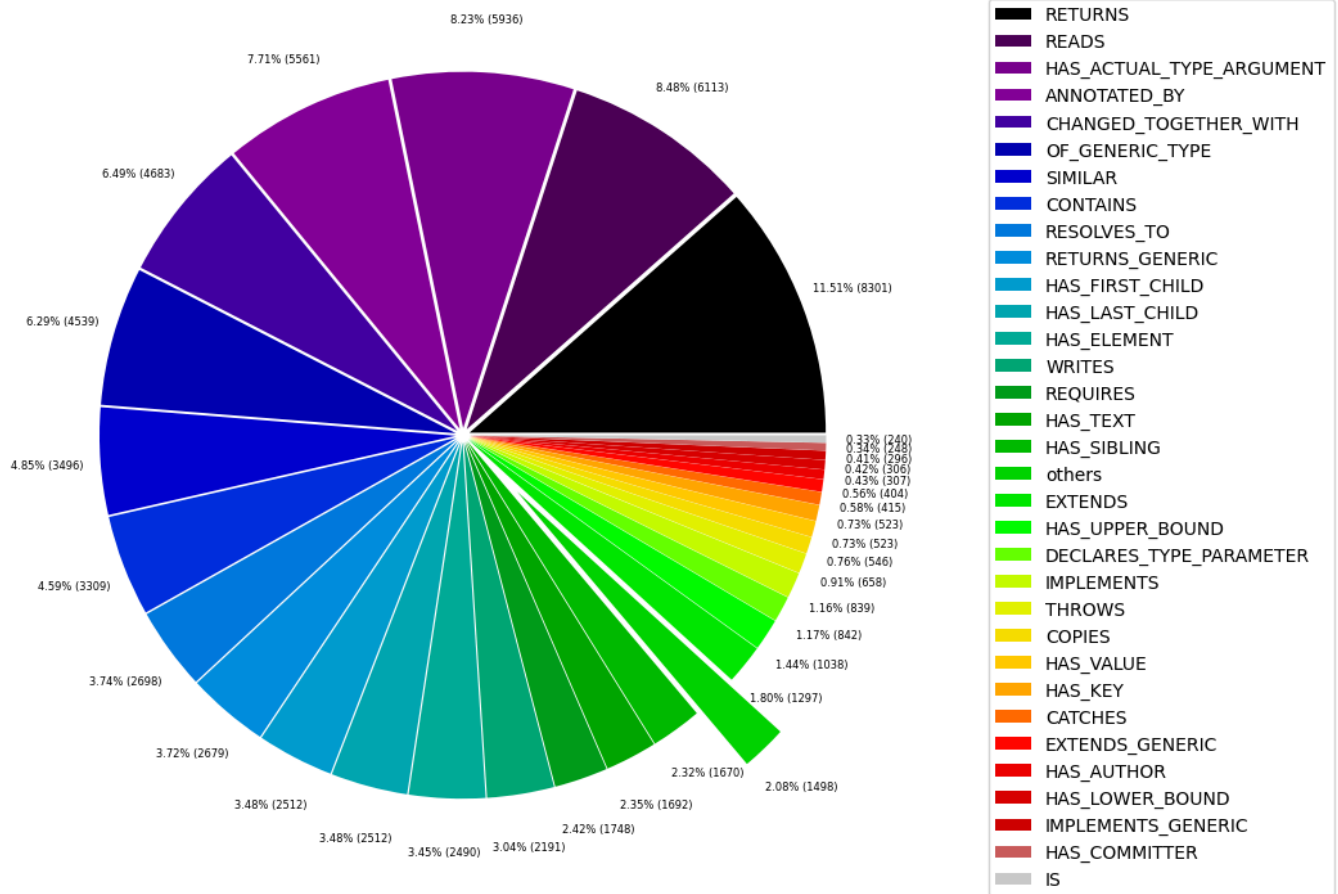
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000060
1	HAS_BRANCH	1	0.000060
2	HAS_HEAD	2	0.000120
3	THROWS_GENERIC	5	0.000300
4	EXCLUDES	6	0.000360
5	HAS_SCM	11	0.000659
6	DESCRIBES	11	0.000659
7	HAS_GOAL	13	0.000779
8	HAS_EXECUTION	13	0.000779
9	HAS_ROOT_ELEMENT	15	0.000899
10	INCLUDES_CONCEPT	19	0.001139
11	IS_ARTIFACT	20	0.001199
12	HAS_CONFIGURATION	20	0.001199
13	USES_PLUGIN	20	0.001199
14	OF_NAMESPACE	22	0.001318
15	HAS_ATTRIBUTE	22	0.001318
16	REQUIRES_TYPE_PARAMETER	26	0.001558
17	REQUIRES_CONCEPT	28	0.001678
18	DECLARES_NAMESPACE	44	0.002637
19	HAS_DEFAULT	54	0.003236
20	HAS_COMPONENT_TYPE	109	0.006533
21	CONTAINS_VALUE	126	0.007551
22	HAS_TAG	147	0.008810
23	ON_COMMIT	147	0.008810
24	TO_ARTIFACT	205	0.012286
25	DECLARES_DEPENDENCY	205	0.012286
26	COPY_OF	206	0.012346
27	IS	240	0.014384
28	HAS_COMMITTER	248	0.014863
29	IMPLEMENTS_GENERIC	296	0.017740

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberO
0	[Git, Update, Change]	UPDATES	[File, Git]	264852	264852	
1	[Git, Update, Change]	MODIFIES	[File, Git]	264852	264852	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	264852	16390	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	99583	16390	
4	[Git, Change, Create]	CREATES	[File, Git]	99583	99583	
5	[Git, Change, Create]	MODIFIES	[File, Git]	99583	99583	
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	61316	16390	
7	[Git, Change, Delete]	MODIFIES	[File, Git]	61316	61316	
8	[Git, Change, Delete]	DELETES	[File, Git]	61316	61316	
9	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	24820	16390	
10	[Git, Change, Rename]	MODIFIES	[File, Git]	24820	24820	
11	[Git, Change, Rename]	DELETES	[File, Git]	24820	24820	
12	[Git, Change, Rename]	CREATES	[File, Git]	24820	24820	
13	[Git, Change, Rename]	RENAMES	[File, Git]	24820	24820	
14	[Git, Commit]	HAS_PARENT	[Git, Commit]	19895	16390	
15	[Repository, File, Git]	HAS_FILE	[File, Git]	16609	1	
16	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	16390	1	
17	[Author, Git, Person]	COMMITTED	[Git, Commit]	16390	307	
18	[Committer, Git, Person]	COMMITTED	[Git, Commit]	16390	248	
19	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	13642	9047	
20	[File, Git]	HAS_NEW_NAME	[File, Git]	9599	16609	
21	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	5528	9047	
22	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	5443	9047	
23	[Java, Value, ByteCode, Annotation]	OF_TYPE	[Type, File, Java, ByteCode, ExternalAnnotatio...	5201	5562	
24	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, javaType]	4218	9457	
25	[Java, ByteCode, Parameter]	ANNOTATED_BY	[Java, Value, ByteCode, Annotation]	3915	9457	
26	[File, Git]	CHANGED_TOGETHER_WITH	[File, Git]	3885	16609	
27	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, javaType]	2884	5029	
28	[Xml, Element]	HAS_ELEMENT	[Xml, Element]	2490	2512	
29	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, javaType]	2376	6545	

Graph Density

```
total_number_of_nodes (vertices): 537827
total_number_of_relationships (edges): 1668574
-> total directed graph density: 5.768475836560317e-06
-> total directed graph density in percent: 0.0005768475836560317
```